

Designing WIRELESS GAS DETECTION And MONITORING SYSTEMS

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Gas detector systems are mainly used for sensing the presence of gas leaks and then taking an action, such as automatically shutting down the source of gas or sounding an alarm to inform the people in the area about the gas leak.

Presented here is a microcontroller (MCU)-based wireless liquefied petroleum gas (LPG) detection and monitoring system. It is multi-node and, therefore suitable for use in caravan sites, holiday homes, students' halls of residence, old peoples' homes, university campuses and other places where groups of people live, and a large number of detectors may be required.

Semiconductor gas sensors

The MQ series of gas sensors are highly popular and are used in most

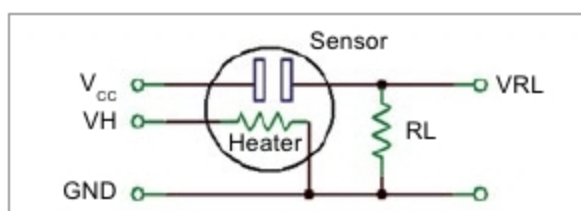


Fig. 1: Connection diagram of MQ series of gas sensors

domestic and industrial gas detection applications. These are semiconductor sensors with SnO_2 as the sensitive material. The sensors have low conductance in clean air, but their conductivity rises when subjected to the gas being sensed. Change in conductivity is converted to an analogue output signal through a simple circuit. Detection ranges



Fig. 3: MQ-6 LPG gas sensor

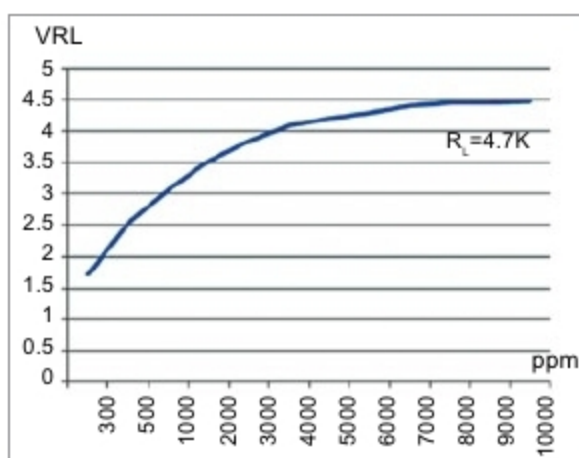


Fig. 4: Output voltage of sensor

of these sensors depend on the type of sensors used.

As shown in Fig. 1, MQ series of gas sensors require two voltage inputs: heater voltage (VH) and circuit voltage (V_{CC}). Heater voltage is specified as 5V and can be AC or DC. Circuit voltage is 5V DC and should be connected as shown in the connection diagram. In most applications, the same 5V voltage is used to power both the heater and the internal sensor circuitry.

The manufacturer provides sensitivity curves showing the gas concentration against the ratio of sensor resistance to resistance in clean air.

MQ-6. This sensor is used to detect LPG concentrations in the air, which are mainly composed of propane and butane. SnO_2 is the sensitive material of this sensor, and its conductivity changes when exposed to LPG.



Fig. 6: TX2H UHF transmitter module

The sensor can detect gas concentrations in the range of 200ppm to 10,000ppm. As with other sensors, an internal

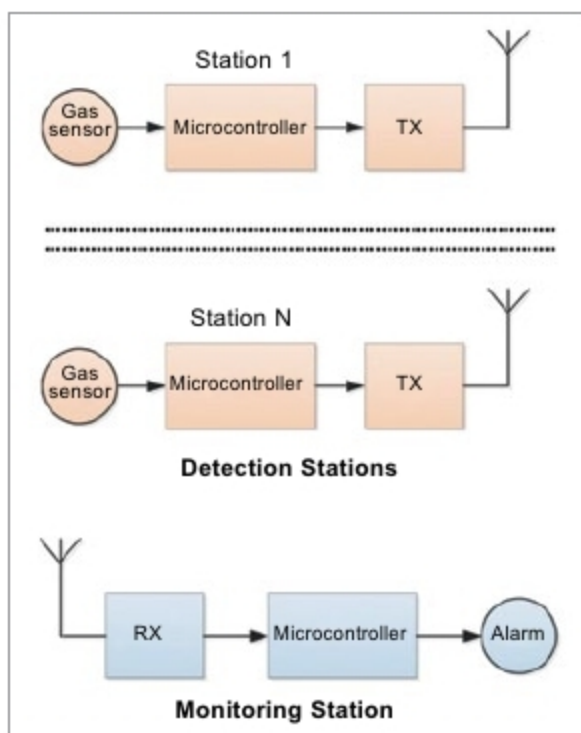


Fig. 2: Block diagram of gas detection system

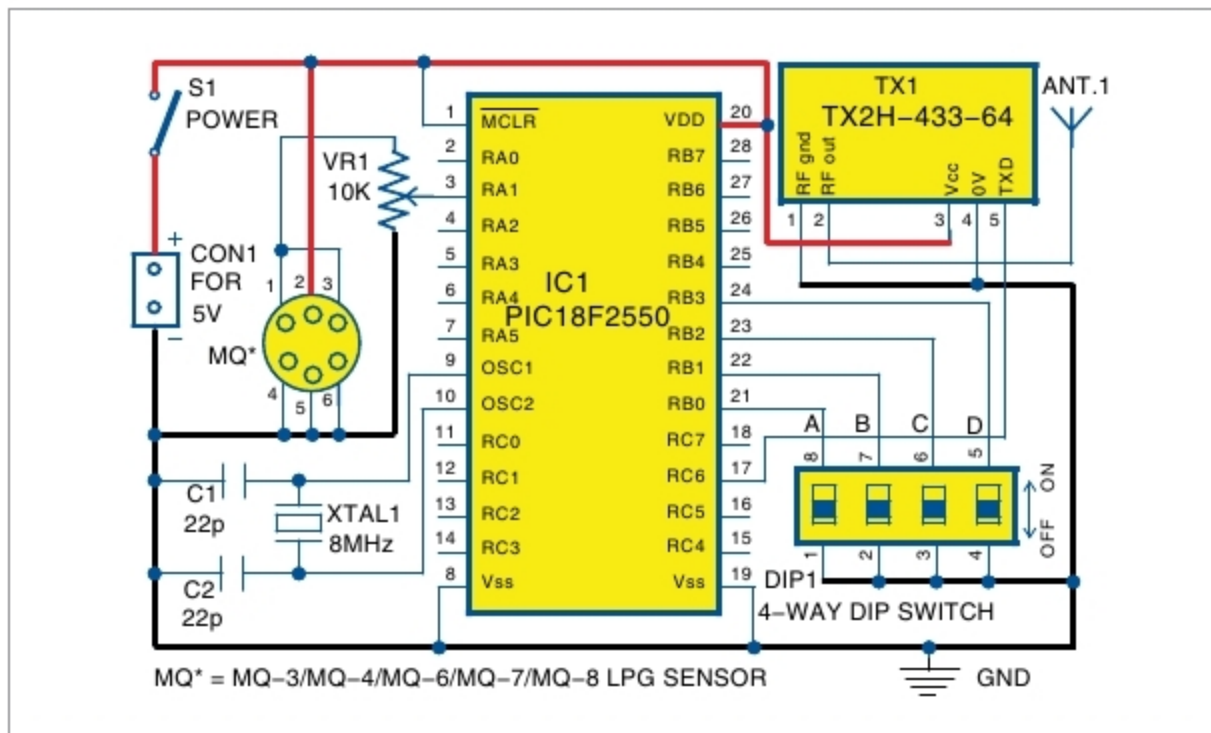


Fig. 5: Circuit diagram of gas detection circuit